

KUMAMOTO, Japan

WMO: 478190

Lat: 32.8136N

Lon: 130.7069E

Elev: 129

StdP: 14.63

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	29.1	31.1	18.1	13.7	40.0	20.5	15.4	39.7	16.7	43.1	15.0	43.7	3.7	10	0.372

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.0	94.5	77.8	92.7	77.4	90.8	76.8	80.0	88.9	79.2	87.6	78.5	86.6	8.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.0	145.9	83.8	77.1	141.9	83.2	76.3	138.1	82.8	43.8	89.0	42.9	88.0	42.2	86.7	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.1	13.7	11.9	24.6	97.3	2.2	1.6	23.0	98.5	21.7	99.4	20.5	100.4	18.9	101.5
			23.2	81.4	2.2	1.1	21.6	82.2	20.3	82.8	19.0	83.5	17.4	84.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	63.6	42.7	45.4	51.5	60.7	69.5	75.3	82.5	84.2	78.1	68.1	56.7	46.6	(6)
(7)		DBStd	15.12	4.94	6.12	5.81	5.60	4.08	3.77	3.74	3.15	4.64	5.40	6.32	5.84	(7)
(8)		HDD50	594	232	156	51	2	0	0	0	0	0	0	13	140	(8)
(9)		HDD65	2669	691	548	418	148	9	0	0	0	0	29	256	570	(9)
(10)		CDD50	5539	6	28	98	324	604	760	1006	1060	843	561	214	35	(10)
(11)		CDD65	2139	0	0	0	20	148	310	541	595	393	125	7	0	(11)
(12)		CDH74	20091	0	0	3	111	867	2031	5889	7060	3451	661	18	0	(12)
(13)		CDH80	7553	0	0	0	8	158	521	2385	3175	1195	111	0	0	(13)
(14)	Wind	WSAvg	4.9	4.5	4.8	5.2	5.1	5.1	5.1	5.5	5.4	4.9	4.8	4.2	4.3	(14)
(15)	Precipitation	PrecAvg	88.8	2.7	3.8	5.9	6.8	7.8	18.9	16.1	8.6	7.6	3.9	3.8	2.8	(15)
(16)		PrecMax	147.1	6.3	7.6	15.0	11.3	16.4	37.9	37.8	21.9	20.8	8.6	8.3	4.8	(16)
(17)		PrecMin	45.1	0.5	1.5	2.7	3.1	2.2	4.2	2.8	2.1	1.3	0.3	0.6	0.1	(17)
(18)		PrecStd	19.3	1.3	1.7	2.5	2.6	3.5	8.9	8.9	5.0	4.9	2.4	2.0	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.4	68.2	73.3	80.9	87.1	90.1	96.3	97.0	93.9	86.8	76.6	66.1	(19)
(20)		MCWB	54.8	58.4	58.7	63.5	65.8	73.1	78.3	78.4	76.7	72.7	64.5	56.9	(20)	
(21)		2%	DB	57.4	63.4	69.0	77.2	83.9	87.6	94.0	95.1	91.2	83.0	72.9	62.4	(21)
(22)		MCWB	50.6	55.4	56.1	60.9	65.5	71.9	77.8	78.2	75.6	69.3	62.3	54.5	(22)	
(23)		5%	DB	54.7	59.6	65.4	74.3	81.5	85.3	92.2	93.5	89.0	80.3	69.9	59.6	(23)
(24)		MCWB	48.1	52.5	55.1	60.1	64.8	71.1	77.5	77.7	74.6	67.5	60.6	53.1	(24)	
(25)		10%	DB	52.2	56.4	62.4	71.4	79.1	83.1	90.2	91.7	86.6	77.7	67.3	56.9	(25)
(26)		MCWB	46.1	49.6	54.1	59.5	64.1	71.0	77.3	77.3	73.7	66.3	60.0	50.8	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.1	61.6	63.2	69.8	73.7	78.3	81.2	81.3	79.5	75.9	68.4	60.5	(27)
(28)		MCDB	60.3	64.4	66.1	74.0	78.9	84.5	90.9	91.1	87.7	81.3	71.7	62.0	(28)	
(29)		2%	WB	52.9	57.6	60.3	66.8	71.4	77.1	80.0	80.2	78.1	73.4	65.4	56.9	(29)
(30)		MCDB	55.1	61.7	64.6	71.6	76.1	82.3	88.8	89.3	85.9	79.7	69.0	60.0	(30)	
(31)		5%	WB	49.4	54.2	57.8	64.4	69.5	76.0	79.1	79.5	77.1	71.1	63.4	54.0	(31)
(32)		MCDB	53.3	58.5	63.3	69.3	74.6	80.9	87.2	88.3	84.6	77.2	67.8	58.0	(32)	
(33)		10%	WB	46.7	50.6	55.3	62.1	67.6	74.8	78.4	78.8	76.0	68.8	61.1	51.6	(33)
(34)		MCDB	51.4	54.9	61.0	68.0	74.0	79.5	86.1	87.4	83.3	74.7	66.5	56.2	(34)	
(35)	Mean Daily Temperature Range	MDBR	15.1	16.1	17.1	18.0	17.4	13.2	12.7	14.0	14.6	16.5	16.3	15.6	(35)	
(36)		5% DB	MCDBR	18.8	19.8	21.3	22.2	21.7	17.5	16.1	16.5	17.0	19.0	18.4	18.3	(36)
(37)		MCWBR	12.8	13.0	12.1	10.2	8.5	6.5	5.1	5.3	6.1	8.1	10.2	12.1	(37)	
(38)		5% WB	MCDBR	15.9	16.2	16.3	14.4	13.5	10.3	14.0	14.3	13.8	13.4	13.7	14.8	(38)
(39)		MCWBR	13.0	13.0	12.0	9.2	8.2	5.2	5.1	5.1	5.5	6.9	9.3	11.6	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.422	0.461	0.521	0.540	0.551	0.559	0.555	0.529	0.500	0.454	0.421	0.413	(40)	
(41)		taud	2.086	2.018	1.868	1.875	1.916	1.976	2.029	2.093	2.136	2.179	2.182	2.128	(41)	
(42)		Ebn at Noon	242	246	241	243	242	239	239	243	245	246	242	237	(42)	
(43)		Edh at Noon	42	49	62	63	62	58	55	50	46	41	37	38	(43)	
(44)	All-Sky Solar Radiation	RadAvg	752	964	1287	1522	1621	1334	1551	1616	1338	1132	854	695	(44)	
(45)		RadStd	75	86	82	140	182	202	211	153	146	121	85	51	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)
(47)	Regional (34 neighbors)	(m) N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page